MIL-B-49461(ER)
AMENDMENT 3

10 July 1987

SUPERSEDING
Amendment 2
29 May 1986

MILITARY SPECIFICATION
BATTERIES, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE

This amendment forms a part of MIL-B-49461(ER), dated 3 March 1986, and is approved for use within the Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

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Add: "3.5.5.1 High Temperature Switch Location. The high temperature switch or thermal fuse shall be located as close as possible to the geometric center of the battery or as indicated below. For a two or three cell in-line arrangement, the high temperature switch or thermal fuse shall be located between any two adjacent cells. For a cluster arrangement of three or more cells, the high temperature switch or thermal fuse shall be located within the cluster."

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3.5.6.2, delete and substitute:
"3.5.6.2 Non-metallic jackets. Non-metallic plastic jackets shall not support combustion nor emit toxic vapors when subjected to flame. The jacket shall not open at any of its seams after being subjected to any test to which the battery is subjected (see 4.8.1)."

3.5.6.2.1. Jacket integrity (paperboard), delete in its entirety.

3.5.6.3, second line, delete "on" and substitute "one".

AMSC N/A

FSC 6135

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3.8.1, delete "Low temperature discharge, (4.8.13.1)" and substitute "Cell discharge, (4.8.13.1)."

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Add the following paragraph:

"3.17 Flame test. Each cell shall contain a pressure relief vent so that when the cell is heated by a direct flame no metal shrapnel from the cell can will be formed and/or ejected upon further cell heating, i.e. the cell can will remain in one piece (see 4.8.20)."

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Table II, In-Process Inspection, add the following:

Inspection	Requirements Paragraph	Test Method Paragraph	Cell	Sampling Plan
Flame Test	3.17	4.8.20	X	3 cells per cell lot

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* 4.7.1.5.1 Delete in its entirety and substitute:

4.7.1.5.1 Sampling plan. A sample of n_s batteries shall be selected at random from production for each shipment lot in amounts determined from the following formula, and rounded off in the case of fractions to an adjacent integer (up or down for each shipment lot), so that exactly n batteries have been assigned to both T and LT capacity test when the sample for the final shipment of the contract lot has been drawn.

$$n_s = a + \frac{N}{N_s}(2n)$$

where:

a = 10 batteries, 5 for ILR capacity test and 5 for D capacity test

n_s = total number of sample batteries to be taken from each shipment lot

N_s = number of batteries in the shipment lot

N = number of batteries in the contract lot (see Table VIII)

n = number of batteries to be taken from the contract lot for each of the two capacity tests, T and LT, in accordance with Table VIII (total number of T and LT samples selected is 2n)

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Insert:

* 4.7.1.5.1.1 Allocation of sample batteries for Group D inspection. The number of batteries n_s, selected from a shipment lot shall be assigned at random at the Government test facility so that 5 batteries will be scheduled for both the ILR and D capacity tests, and the remainder split approximately in half for the T and LT tests. Such assignment shall result in the allocation of exactly n batteries to both T and LT capacity tests after the final shipment on the contract lot is made. If necessary, the sample size n_s taken from the last shipment lot of a contract lot shall be adjusted so that this result is achieved.

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* Table VIII, delete in its entirety and substitute:

Table VIII Sample size and allowable failures for both T and LT capacity tests.

Contract lot size "N"	Sample size "n" for T and Lt tests from contract lot	1/ Acceptance No. for T and LT Test
0 to 110	2/ 5	2/ --
111 to 500	15	3
501 to 800	25	5
801 to 1,300	35	7
1,301 to 3,200	50	9
3,201 to 8,000	75	13
8,001 to 22,000	110	18
22,001 to 110,000	150	24
over 110,000	225	34

1/ When the number of capacity values falling below the minimum requirements specified (see 3.1) for a given test is equal to or less than the associated acceptance number, the contract lot from which the sample was drawn has met the requirements of that test.

2/ Determination of compliance specified in 4.7.1.5.2 shall not apply to contract-lot sizes of less than 111.

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* 4.7.1.5.2 Delete in its entirety and substitute:

4.7.1.5.2 Compliance. The entire contract lot shall be considered as complying when the T and Lt tests show compliance as specified in Table VIII and the number of ILR failures for the contract lot does not exceed twenty (20) percent of the number of samples subjected to ILR testing.

* 4.7.1.5.3.1 and 4.7.1.5.3.2, delete the term "inspection battery lot" wherever it appears and substitute "contract lot".

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4.8.7 Altitude. Add to end of paragraph "Open circuit voltage only shall be observed after the test."

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4.8.9.1.7, 3rd line, delete "and 130°F"

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4.8.9.1.9, delete "+70 ±°F" and substitute "+ 70 ± 5 °F."

4.8.10 delete and substitute:

"4.8.10 Safety feature. Each sample battery shall be subjected to a direct short, i.e., less than 0.1 ohm, until the circuit is broken by the safety feature. The maximum current and time required to activate the safety feature shall be determined and recorded. Following the direct shorting, all batteries shall be placed at +195°F for two hours. At the end of two hours the battery must be checked to see that no cell has vented. If any cell has vented, the battery has failed this test. If no cell has vented, the ambient temperature shall be raised to a maximum of 320°F. Batteries shall be maintained at an ambient of +320°F for two hours. Any cell that vented at or below 320°F must be checked to insure that the venting occurred only through the designed vent a failure. Batteries containing cells which exploded, burned or did not vent are failures (see 3.5.4). Distortion of labeling or dimension is permitted.

4.8.11.2, Jacket integrity (paperboard); delete in its entirety.

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4.8.13.1, delete and substitute:
"4.8.13.1 Cell discharge test. Following the gas leakage test, the same thirty (30) cells used in the gas leakage test shall be stored at 70°F± 5°F for a minimum of six (6) hours. Each cell shall be discharged at 70°± 5°F through the load and to end voltage in accordance with the individual specification sheet. There shall be at least a two (2) inch separation between all cells that are being discharged. Following capacity determination, the discharge shall continue to zero volts."

4.8.17, First line, delete "ration" and substitute "ratio." Second line, delete "imspection" and substitute "inspection."

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Add: "4.8.20 Flame test. This test shall be performed on cells prior to installation in batteries. Physically wire cell to lay horizontally above a torch. Prior to the flame test the cell shall be discharged 75% of its capacity. Apply flame of sufficient heat to cause the cell to vent in three minutes maximum. Keep flame and cell in the same position until a lithium fire occurs (see 3.17).

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